

Multiple Choice

1. **Answer:** C

Options: A) 84 birds; B) 45 birds; C) 21 birds; D) 34 birds; E) 42 birds

Note: Correct option: C - 21 birds

2. **Answer:** B

Options: A) 90 miles; B) 120 miles; C) 150 miles; D) 140 miles; E) 26 miles

Note: Correct option: B - 120 miles

3. **Answer:** B

Options: A) $4 - t = 112$; $-\$108$; B) $4t = 112$; $\$28$; C) $t \text{ over } 4 = 112$; $\$448$; D) $t + 4 = 112$; $\$108$; E) $t \text{ over } 4 = 112$; $\$28$

Note: Correct option: B - $4t = 112$; $\$28$

4. **Answer:** A

Options: A) 0.6931471; B) 3.7320508; C) 2.3025851; D) 6.2831853; E) 1.4142135

Note: Correct option: A - 0.6931471

5. **Answer:** A

Options: A) 3; B) 4; C) 1; D) 5; E) 9

Note: Correct option: A - 3

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** We have $\& \sqrt{[3]\{12\}} \times \sqrt{[3]\{20\}} \times \sqrt{[3]\{15\}} \times \sqrt{[3]\{60\}}$
 $\& = \sqrt{[3]\{2^2 \cdot 3^1\}} \times \sqrt{[3]\{2^2 \cdot 5^1\}} \times \sqrt{[3]\{3^1 \cdot 5^1\}} \times \sqrt{[3]\{2^2 \cdot 3^1 \cdot 5^1\}}$
 $\& = \sqrt{[3]\{(2^2 \cdot 3^1)(2^2 \cdot 5^1)(3^1 \cdot 5^1)(2^2 \cdot 3^1 \cdot 5^1)\}}$
 $\& = \sqrt{[3]\{(2^2 \cdot 2^2 \cdot 2^2)(3^1 \cdot 3^1 \cdot 3^1)(5^1 \cdot 5^1 \cdot 5^1)\}}$
 $\& = \sqrt{[3]\{(2^6)(3^3)(5^3)\}}$
 $\& = \sqrt{[3]\{2^6\}} \times \sqrt{[3]\{3^3\}} \times \sqrt{[3]\{5^3\}}$
 $\& = (2^2)(3)(5) = 60.$

Note: Students should show all steps in their mathematical solution.

12. **Answer:** 2×2 . Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.